Phase Correlation Analysis of a Gain-Switched DFB Laser

Effect of repetition rate on pulse-to-pulse intensity and phase coherence
with and without CW SLD injection (1 – 10 GHz, step 1 GHz)

DFB Laser (1550 nm, InGaAsP/InP)

Wavelength	**1550 nm**
Cavity length	**300 µm**
Threshold current	**16.67 mA**
Photon lifetime τ_p	**3.10 ps**
Henry α factor	**3.0**
Diff. gain a	**2.50e-20 m²**
Gain compression ε	**3.00e-23 m³**
Spont. coupling β	**1.0e-04**

SLD Injection (CW, broadband)

SLD wavelength	**1550 nm**
SLD current	**150 mA**
SLD output power	**6.69 mW**
Coupling efficiency η	**0.10**
Injection rate κ	**1.114e+10 s⁻¹**
Injected density S_inj	**4.310e+17 m⁻³**
Frequency detuning Δν	**0 GHz**

Simulation Parameters

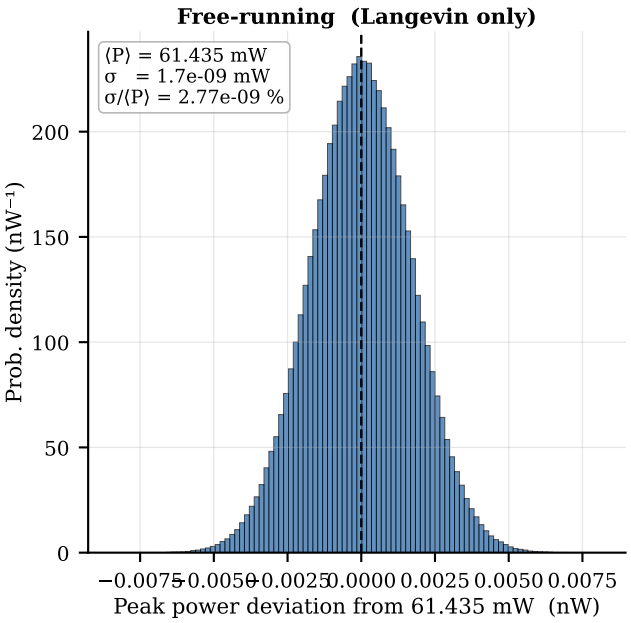
Integration method	**Euler-Maruyama (SDE)**
Time step dt	**2 ps**
Parallel trains K	**200**
Pulses per train	**5,000**
Total pulses	**1,000,000**
Bias I_bias	**0.9 × I_th**
Peak I_peak	**5.0 × I_th**
Rise time t_rise	**5% of T_rep**

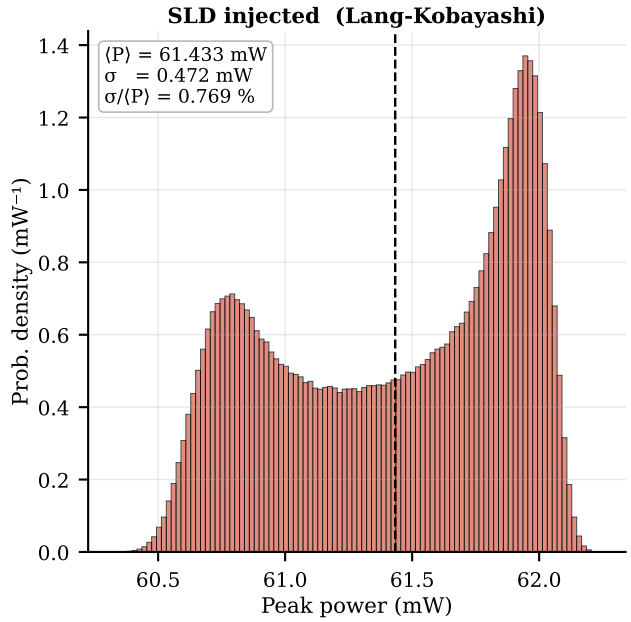
Noise (Langevin / Gardiner)

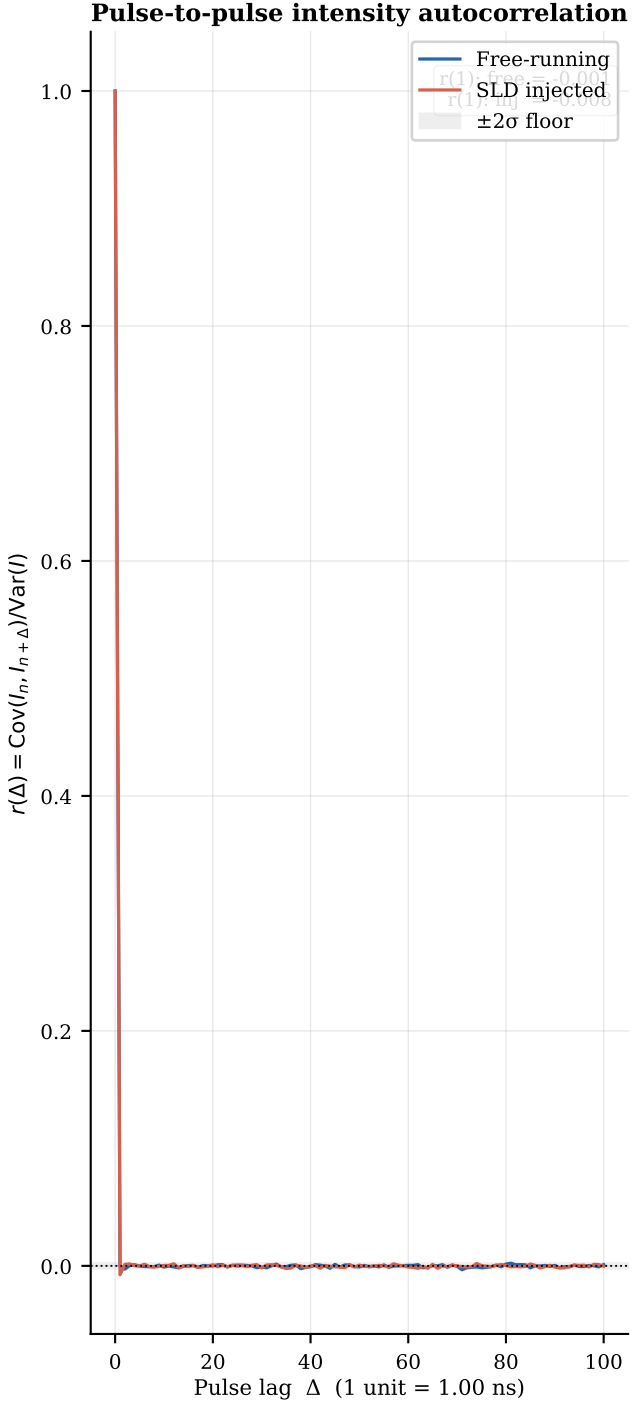
Carrier noise σ_N	**√(2 R_sp) per √s**
Photon noise σ_S	**√(2 β B N²) per √s**
Phase noise σ_φ	**√(β B N² / 2S) per √s**
Injection (free-run)	**κ = 0, S_inj = 0**
Injection (seeded)	**κ as above, S_inj as above**
Lang-Kobayashi term	**2κ√(S·S_inj) cos φ in dS/dt**

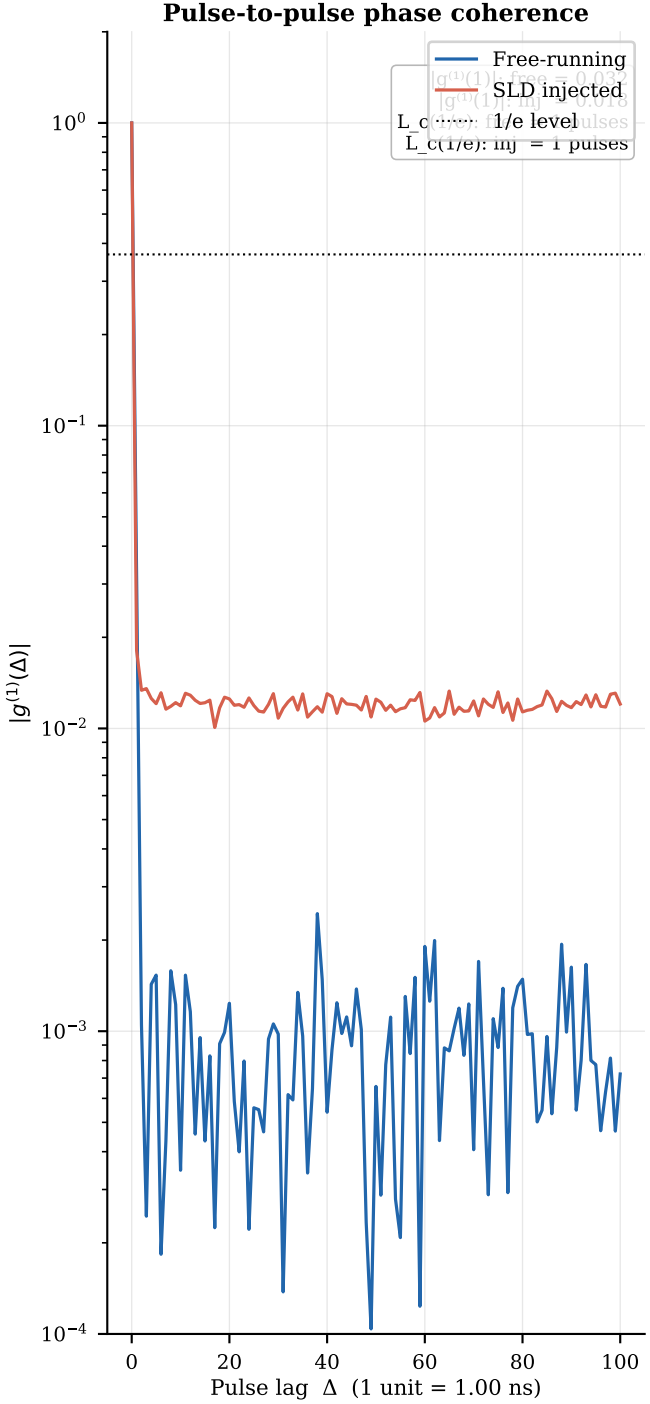
Sampling: photon density S and accumulated phase φ are recorded at the pulse-intensity peak (located via a deterministic burn-in) for each of the 1,000,000 simulated pulses.

Repetition rate f_rep = 1 GHz (T_rep = 1.00 ns) N = 1,000,000 pulses sampled at t = 162.0 ps into each pulse

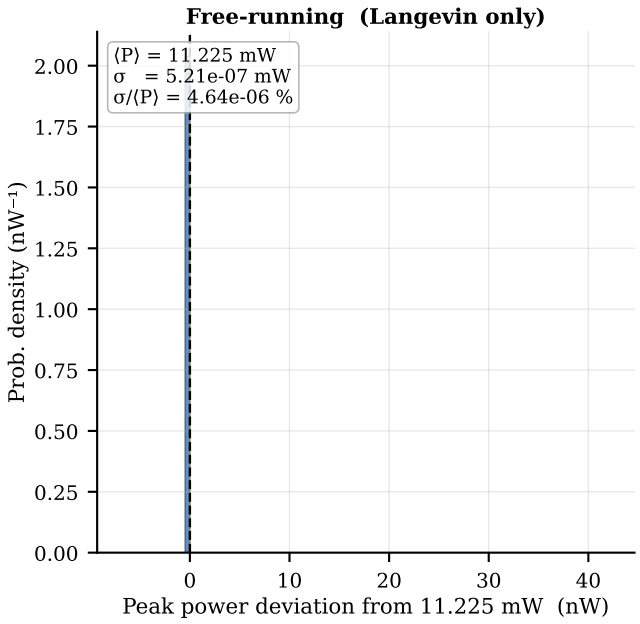


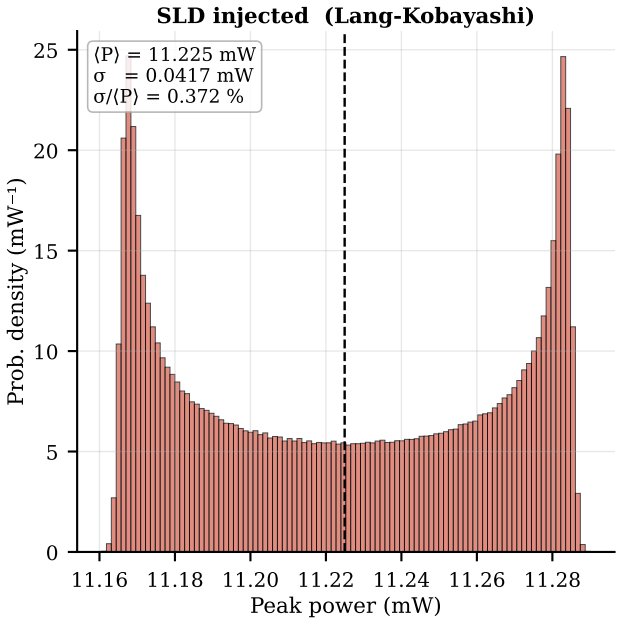


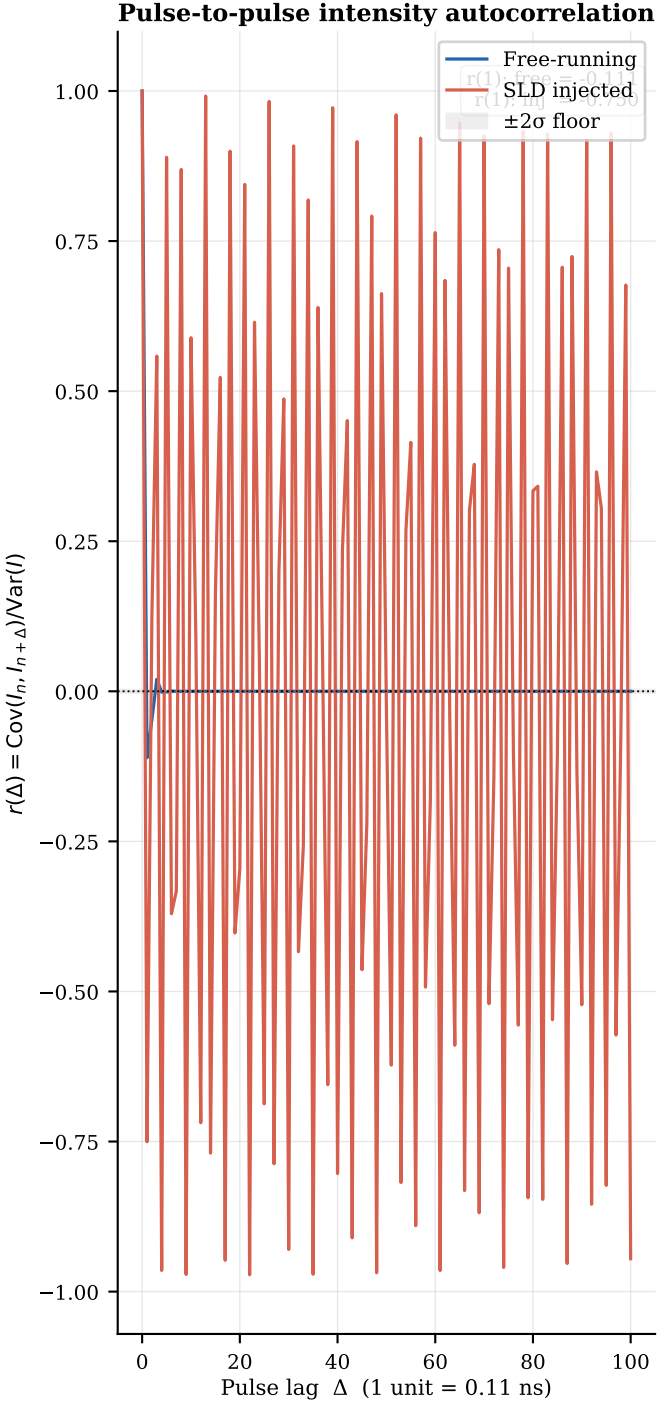


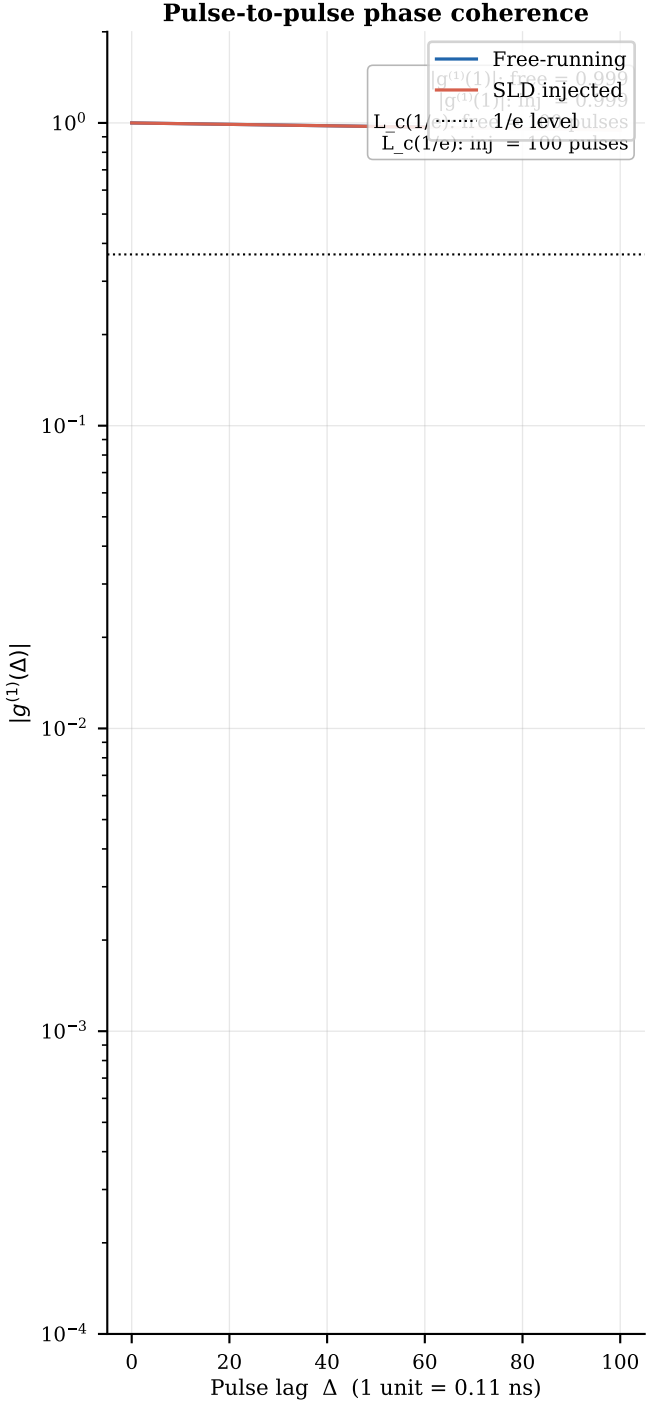


Repetition rate f_rep = 9 GHz (T_rep = 0.11 ns) N = 1,000,000 pulses sampled at t = 60.0 ps into each pulse

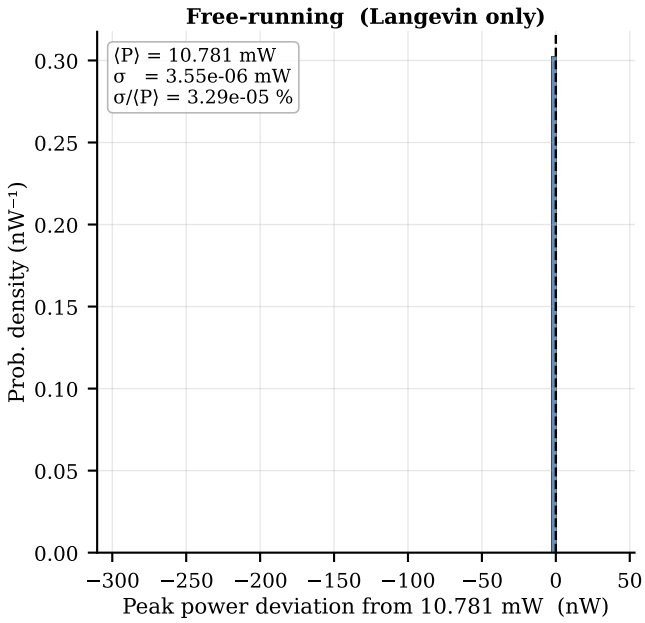


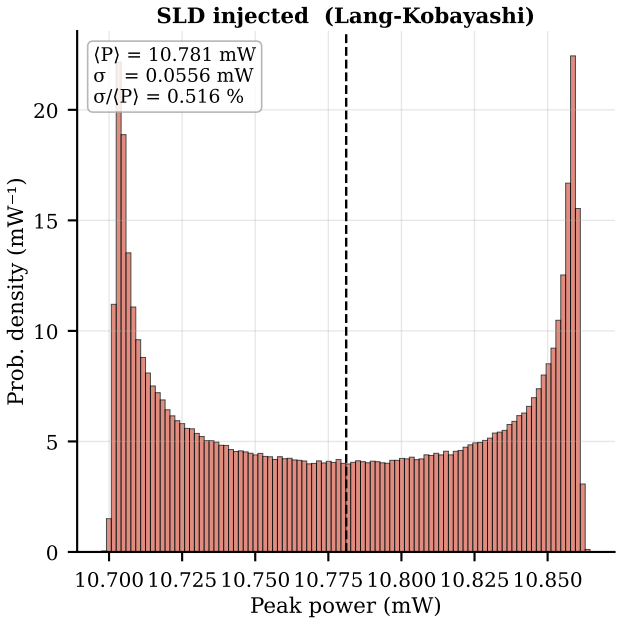


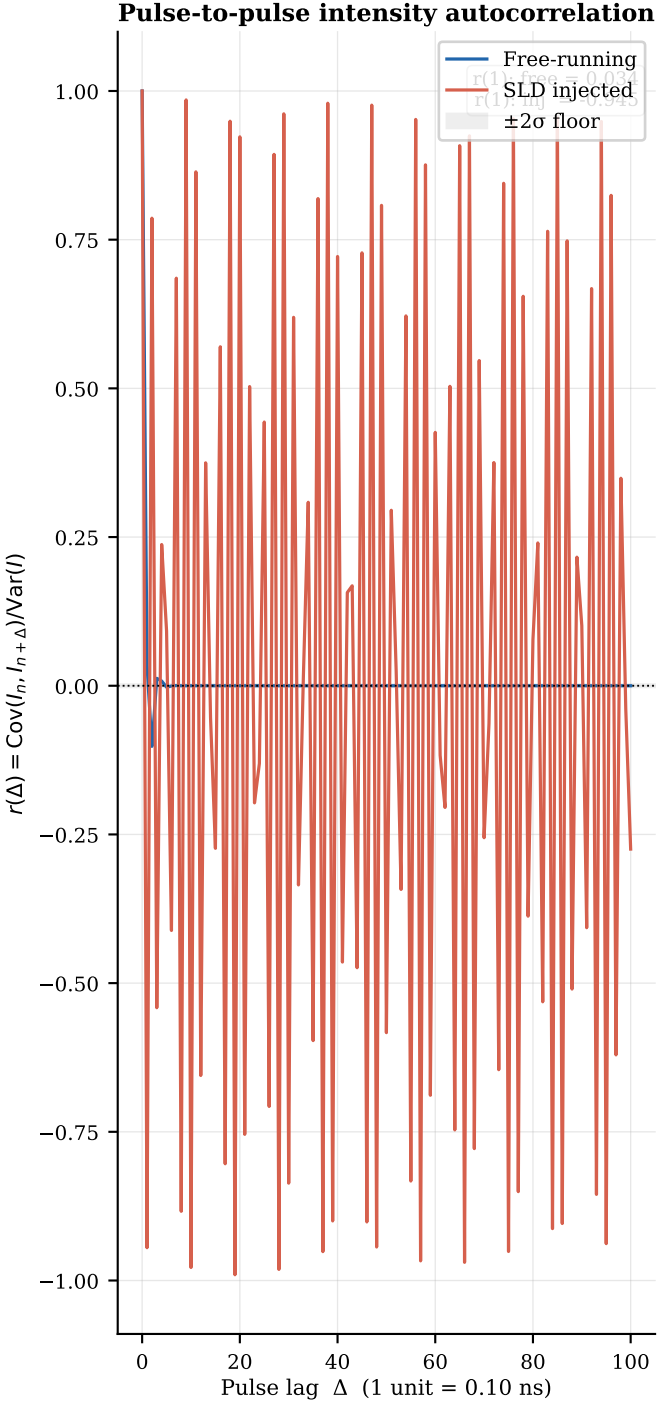


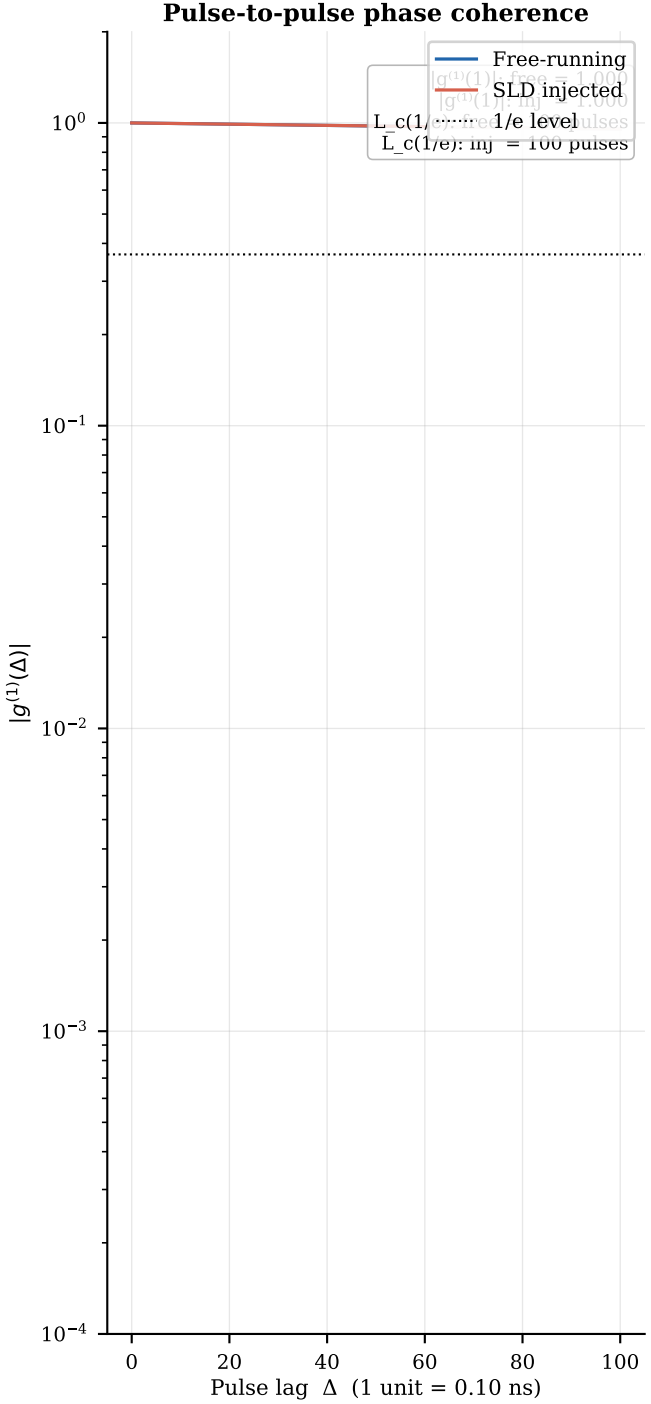


Repetition rate f_rep = 10 GHz (T_rep = 0.10 ns) N = 1,000,000 pulses sampled at t = 56.0 ps into each pulse

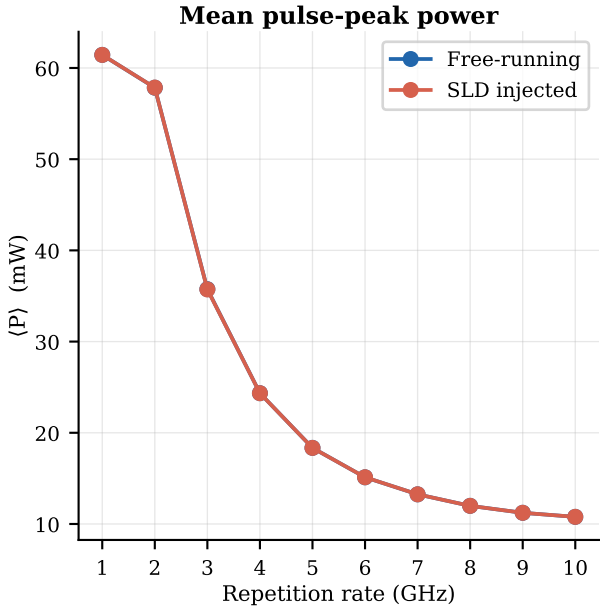


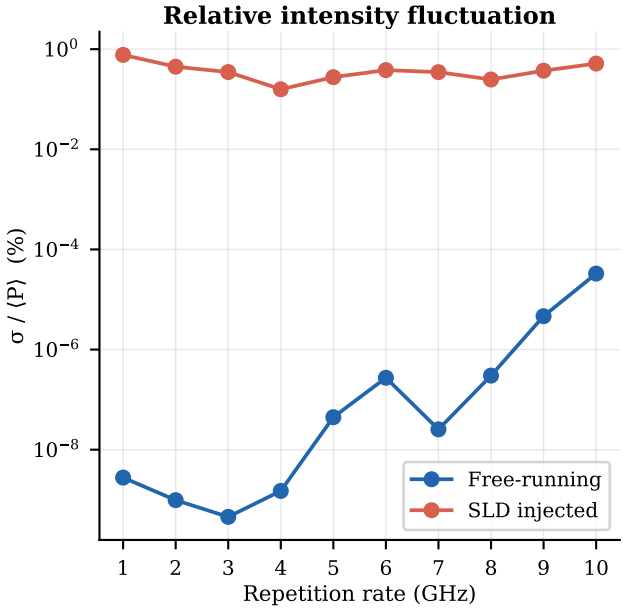


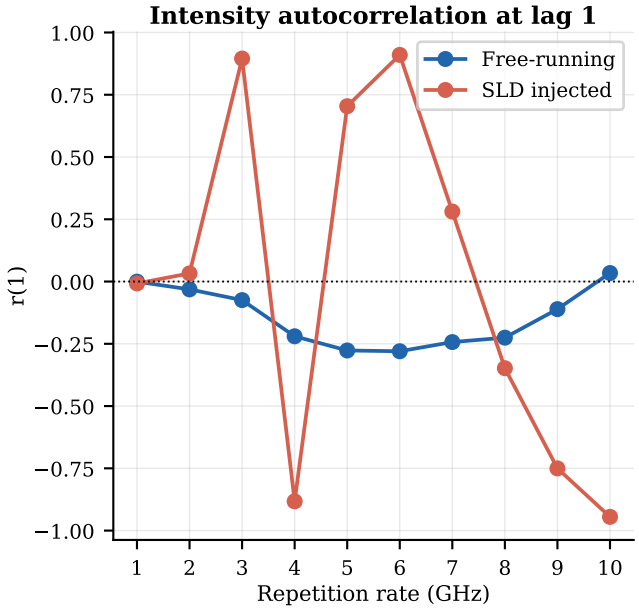


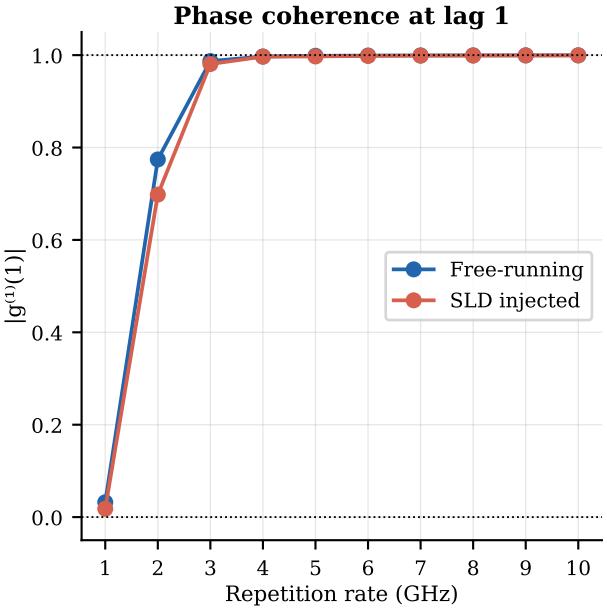


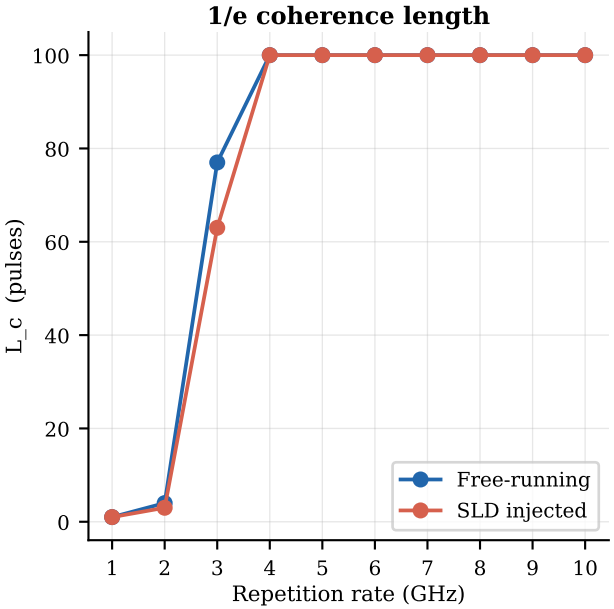
Summary: Pulse-to-Pulse Statistics vs Repetition Rate (1 – 10 GHz)











Phase coherence summary

f (GHz)	$\|g^{(1)}(1)\|$ free	$\|g^{(1)}(1)\|$ inj	L_c free	L_c inj
1	0.032	0.018	1	1
2	0.774	0.698	4	3
3	0.987	0.981	77	63
4	0.997	0.997	100	100
5	0.999	0.997	100	100
6	0.999	0.998	100	100
7	0.999	0.999	100	100
8	0.999	0.999	100	100
9	0.999	0.999	100	100
10	1.000	1.000	100	100